

SAGAR P

INFORMATION SCIENCE & ENGINEERING UNDERGRADUATE • FULL-STACK & SYSTEMS DEVELOPER

SUMMARY

Motivated Information Science & Engineering undergraduate at Atria Institute of Technology (CGPA 9.3) with solid foundations in Python, full-stack application development, and algorithmic optimization. Proven project lead in building mathematical engines (Linear Diophantine cash optimization) and production academic platforms; active student organizer with the Cultural Committee Atria managing campus logistics and audio systems. Eager to contribute technical execution, rapid prototyping, and collaborative energy to the GDG on Campus Core Team.

PROJECTS

ATM Cash Combination Finder

Project Lead • Oct 2024

Python • Flask • SQLite • Linear Diophantine Math • Backtracking

[github.com/sagar-core/ATM-Cash-Finder]

- Formulated and implemented an algorithm resolving Linear Diophantine equations to compute all valid dispensing combinations across ₹500, ₹200, ₹100, and ₹50 currency denominations.
- Engineered optimization heuristics prioritizing minimal physical bill count while enforcing real-time currency cassette limits to prevent premature denomination depletion.
- Built a responsive customer portal with digital withdrawal receipts and a secure administrative dashboard for cash restocking; presented mathematical proof directly to the Mathematics HOD.

Student-Teacher Appointment Platform

Backend & Core Architecture

Python • Flask • MySQL • JavaScript • HTML5/CSS3

- Designed an academic consultation scheduling platform eliminating student walk-in conflicts and automating faculty office-hour bookings across departments.
- Built role-based authentication workflows, dynamic slot availability calendars, approval dashboards, and automated email confirmation notifications.

Donor Relationship Management (CRM) Prototype

Centre for Wildlife Studies (CWS)

Python • Structured Schemas • JavaScript • AI Integration Heuristics

- Architected a centralized CRM prototype for wildlife conservation operations, replacing dispersed spreadsheets with unified donor histories, pledge tiers, and campaign logs.
- Integrated AI-assisted reminder workflows generating contextual follow-up prompts and impact report drafts to enhance long-term supporter engagement.

EXPERIENCE & LEADERSHIP

Event Operations & Logistics Volunteer

2024 – Present

Cultural Committee Atria (CCA) • Atria Institute of Technology

- Co-organized college-wide flagship events (Community Day), managing 15+ student stalls, vendor coordination, and crowd logistics for 500+ attendees.
- Managed technical audio systems and stage setups, monitoring live sound levels and AV cues during college ceremonies and hackathon presentations.
- Coordinated communications with faculty leads and external participants to ensure smooth registration pipelines and operational execution.

HACKATHONS & ACHIEVEMENTS

- 3rd Place — Atria Internal Hackathon (SIH Selection Round):** Competed among ~30 competitive teams; led frontend interface design, structured data handling, and the final technical pitch.
- Departmental HOD Presentation:** Invited to demonstrate the ATM Cash Combination Optimizer's mathematical models to the Mathematics Department Head.
- Pre-University Merit Scholarship:** Awarded scholarship recognition for achieving **92% Distinction** in 10th-grade board examinations.
- Academic Excellence Award:** Conferred distinction honours for achieving **90% Distinction** in 12th-grade Pre-University Board exams.

CONTACT

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SKILLS

- Programming:** Python, C++, Java, SQL
- Web & Backend:** Flask, REST APIs, HTML5, CSS3, JavaScript (ES6+)
- Databases:** SQLite, MySQL
- Tools & VCS:** Git, GitHub, VS Code
- AI & APIs:** Gemini API, Prompt Design
- Core Concepts:** Data Structures, OOP, Backtracking Heuristics
- Operations:** Event Logistics, Live Audio & Stage AV Systems

EDUCATION

B.E. in Information Science & Engineering

Atria Institute of Technology, Bengaluru
2025 – 2029

CGPA: 9.3

Pre-University Course (PCMC)

Seshadripuram PU College, Bengaluru
Distinction • Academic Award

90%

Secondary School Certificate (SSLC)

Sathya English High School, Bengaluru
Distinction • Merit Scholarship

92%

AREAS OF FOCUS

- Full-Stack Web Development
- Mathematical & Algorithmic Optimization
- Developer Community & Event Organizing
- Rapid Prototyping & AI Workflow Tooling

LANGUAGES

English	Professional Working
Kannada	Native / Fluent
Hindi	Working Proficiency